

Problem set 6

Logistic Regression

Exercise 1

1. Let $\overline{W}^T = (w_1, w_2, \dots, w_d)$ and $\overline{P}^T = (p_1, p_2, \dots, p_d)$ be points from R^d and let b be a number from R . Compute the distance from $\overline{P}$ to the hyperplane defined by the equation $\overline{W}^T \overline{X} + b = 0$.

Exercise 2

Show the following properties of the logistic sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$.

1. $\sigma(x) = 1 - \sigma(-x)$
2. $\frac{\partial \sigma}{\partial x} = e^{-x} \sigma^2(x)$

Exercise 3

If the probability of an event is $\frac{1}{2}$, what are the odds? What about the log odds? What does this tell us about the qualitative interpretation of odds and log odds?

Exercise 4

Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\beta_0 = -6$, $\beta_1 = 0.05$, $\beta_2 = 1$.

1. Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.
2. How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?